

Minimax-rate optimality of MONK under adversarial contamination

A literature-implied answer to the question in arXiv:1802.04784

Autonomous mathematical-discovery run `fa_banach_001`

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Status: literature-implied answer (full minimax-rate resolution). The order of MONK’s error due to adversarial outliers is optimal under the paper’s finite-second-moment hypothesis. The relation to MONK is identified here; the supporting authors do not explicitly state an RKHS corollary. Exact leading constants and the exact breakdown behavior as the fraction of contaminated blocks approaches one half are not claimed.

Source question and supporting result

The source is M. Lerasle, Z. Szabó, T. Mathieu, and G. Lecué, *MONK—Outlier-Robust Mean Embedding Estimation by Median-of-Means*, arXiv:1802.04784, ICML 2019. In the published PDF, page 4, “Remarks,” bullet “Dependence on δ ,” the authors ask:

Whether this robustness to outliers is optimal in the studied setting is an open question.

Their Theorems 1–2 and the detailed proof in Appendix A give the upper bound needed below.

The decisive later source is X. de Juan and S. Mazuelas, *On the Optimality of the Median-of-Means Estimator under Adversarial Contamination*, arXiv:2510.07867, NeurIPS 2025. Its equation (2) and Theorem 3.2 identify

$$\sigma \left(\sqrt{\frac{\log(2/\eta)}{N}} + \sqrt{\alpha} \right)$$

as the finite-variance minimax order and prove that scalar median-of-means, with the number of blocks of order $\log(1/\eta) + \alpha N$, attains it. That paper discusses scalar means, not the MONK min-max estimator. The connection below is therefore an implication rather than an explicit answer.

Rate-level answer

Let $Z = K(\cdot, X)$ take values in a separable RKHS $\mathcal{H}$, let $\mu = \mathbb{E}Z$, and let $\Sigma = \mathbb{E}[(Z - \mu) \otimes (Z - \mu)]$ be trace class. From N clean i.i.d. observations, an adversary may inspect the sample and replace at most m entries. The same convention is used for m contaminated pairs in the two-sample MMD problem.

Theorem 1 (MONK is optimal in its outlier dependence). *There are universal constants $c, C > 0$ with the following property. If $0 < \eta \leq 1/4$ and*

$$m + \log(1/\eta) \leq cN,$$

then MONK can be formed with $Q \asymp m + \log(1/\eta)$ blocks so that, with probability at least $1 - \eta$,

$$\|\widehat{\mu}_{P,Q} - \mu_P\|_{\mathcal{H}} \leq C \max \left\{ \sqrt{\frac{\text{Tr } \Sigma_P}{N}}, \sqrt{\frac{\|\Sigma_P\| [m + \log(1/\eta)]}{N}} \right\}. \quad (1)$$

Likewise the MONK MMD estimator satisfies

$$|\widehat{\text{MMD}}_Q(P, R) - \text{MMD}(P, R)| \leq C \max \left\{ \sqrt{\frac{\text{Tr } \Sigma_P + \text{Tr } \Sigma_R}{N}}, \sqrt{\frac{(\|\Sigma_P\| + \|\Sigma_R\|)[m + \log(1/\eta)]}{N}} \right\}. \quad (2)$$

The corruption term in (1), and also that in (2), cannot be improved in order over the class of distributions with only a finite second moment. More precisely, for every $1 \leq m \leq N/4$ and every estimator, there is already a one-dimensional linear-kernel instance of covariance at most σ^2 and an adversary replacing at most m observations for which an error

$$c\sigma\sqrt{m/N} \quad (3)$$

occurs with a universal positive probability.

Thus the answer to the source question is affirmative in the standard minimax-rate sense: under the stated trace-class covariance assumption, MONK has the optimal $\sqrt{m/N}$ outlier dependence. The clean-sample and confidence terms were already known to be optimal in the source paper's discussion.

Why the implication holds

Upper bound. The detailed proof of Theorem 1 in arXiv:1802.04784 establishes, before its confidence reparameterization, that whenever $m \leq Q(1/2 - \gamma)$,

$$\|\widehat{\mu}_{P,Q} - \mu_P\|_{\mathcal{H}} \leq C_0 \max \left\{ \sqrt{\frac{\|\Sigma_P\|Q}{\gamma N}}, \frac{1}{\gamma} \sqrt{\frac{\text{Tr } \Sigma_P}{N}} \right\} \quad (4)$$

with probability at least $1 - \exp(-Q\gamma^2/72)$. Choose $\gamma = 1/4$ and an integer

$$Q \geq \max\{4m + 1, \lceil 1152 \log(1/\eta) \rceil\}$$

within a universal factor of the right-hand side. Equation (4) becomes (1). If Q does not divide N , discard fewer than Q observations and use $N' = Q \lfloor N/Q \rfloor$; under the displayed smallness condition, $N' \asymp N$, and at most m retained observations are still corrupted. The Appendix A proof of Theorem 2 gives the identical substitution with the sums of the two covariance norms and traces, proving (2).

This is exactly the block-count balance isolated for scalar median-of-means in Theorem 3.2 of arXiv:2510.07867. The source MONK proof supplies the dimension-free Hilbert-space lift.

Matching lower bound for mean embeddings. It is useful to record a self-contained reduction. Take $\mathcal{X} = \mathbb{R}$ and the continuous positive-definite linear kernel $K(x, y) = xy$. Its RKHS is $\mathbb{R}$, its feature map is x , and its kernel mean embedding is the ordinary scalar mean. Put

$$\rho = \frac{m}{2N}, \quad A = \frac{\sigma}{\sqrt{\rho(1-\rho)}}, \quad P_{\pm} = (1 - \rho)\delta_0 + \rho\delta_{\pm A}.$$

Both alternatives have variance exactly σ^2 and means

$$\mu_{\pm} = \pm \sigma \sqrt{\frac{\rho}{1-\rho}}.$$

Let J be the number of nonzero observations. Since $\mathbb{E}J = N\rho = m/2$, Markov’s inequality gives $\Pr(J \leq m) \geq 1/2$. After inspecting the sample, the adversary replaces all nonzero observations by zero whenever $J \leq m$. On this event the statistician sees the all-zero data set under either sign. For any fixed output on that data set, one of the two signs has error at least

$$\sigma \sqrt{\frac{\rho}{1-\rho}} \geq \sigma \sqrt{\frac{m}{2N}}.$$

This proves (3) with failure probability at least $1/2$ for deterministic estimators; the usual conditioning argument gives a universal constant for randomized estimators.

Matching lower bound for MMD. Fix $R = \delta_0$. Compare $P_0 = \delta_0$ with $P_1 = (1 - \rho)\delta_0 + \rho\delta_A$, with the same ρ, A . For the linear kernel,

$$\text{MMD}(P_0, R) = 0, \quad \text{MMD}(P_1, R) = \sigma \sqrt{\frac{\rho}{1-\rho}}.$$

The observed data are identically all zero under P_0 and, after the same legal attack, with probability at least $1/2$ under P_1 . No estimator can be within half the displayed separation for both alternatives on that common observation. This again yields (3) with universal positive probability.

Scope and provenance

The theorem settles the dependence on the sample corruption fraction in the finite-second-moment model. It does *not* optimize constants, determine the exact finite-sample replacement breakdown point, or prove that the source paper’s stated 25% breakdown value is the largest possible. Those are distinct questions from minimax rate optimality.

A bounded search used the exact MONK title and question wording, “MONK optimal robustness outliers,” and “mean embedding adversarial contamination minimax finite variance.” It found arXiv:1812.03523, which proves optimal contamination exponents under stronger $2 + \delta$ moments, and the decisive arXiv:2510.07867, which proves the exact finite-variance scalar MoM rate. Because the latter makes the core rate result known and the RKHS conclusion is a direct identification using the earlier MONK bound, this packet is not a claim of new mathematics.

References

1. M. Lerasle, Z. Szabó, T. Mathieu, and G. Lecué, *MONK—Outlier-Robust Mean Embedding Estimation by Median-of-Means*, arXiv:1802.04784; Proceedings of ICML 2019, vol. 97, pp. 3782–3793.
2. X. de Juan and S. Mazuelas, *On the Optimality of the Median-of-Means Estimator under Adversarial Contamination*, arXiv:2510.07867; NeurIPS 2025.
3. S. Minsker, *Uniform bounds for robust mean estimators*, arXiv:1812.03523 (background comparison only).